Lesson 4: Data Encryption Standard (DES)

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Lecture Goals

By the end of this session, students should be able to:

- Understand the architecture of a symmetric block cipher.
- Explain the components of the Feistel Network.
- Identify the role of the Initial Permutation, Expansion, and S-boxes.
- Understand the DES key schedule and the "56-bit key" problem.
- Differentiate between Confusion and Diffusion.

1 Introduction

The Data Encryption Standard (DES) is a symmetric-key block cipher published in 1977. It was the first encryption standard widely adopted by the industry and government.

Key parameters:

- **Block Size:** 64 bits.
- **Key Size:** 64 bits (56 bits active + 8 parity bits).
- **Structure:** Feistel Network with 16 rounds.

2 The Feistel Network

The core of DES is the Feistel structure. A 64-bit block is split into two 32-bit halves: Left (L) and Right (R). In each round i:

$$L_i = R_{i-1}$$
$$R_i = L_{i-1} \oplus f(R_{i-1}, K_i)$$

Where f is the **Round Function** and K_i is the 48-bit subkey. A major advantage of this design is that encryption and decryption are practically identical.

3 The Round Function f

The function f transforms the data in four distinct steps:

1. **Expansion (E):** Expands 32-bit R to 48 bits by duplicating specific bits.
2. **Key Mixing ($\oplus$):** XORs the expanded data with the 48-bit subkey K_i.
3. **Substitution (S-boxes):** The only non-linear step. 48 bits are mapped to 32 bits using 8 lookup tables (S-boxes).
4. **Permutation (P):** Shuffles the 32 bits to provide diffusion.

4 Confusion and Diffusion

DES implements two fundamental properties of secure ciphers:

- **Confusion:** Obscures the relationship between the key and ciphertext (via S-boxes).
- **Diffusion:** Spreads the influence of one plaintext bit over many ciphertext bits (via P-box and Feistel swaps).

5 Key Schedule

The 56-bit effective key is used to generate 16 unique 48-bit subkeys. The process involves:

- **PC-1:** Initial permutation and removal of parity bits.
- **Left Circular Shifts:** Rotating bits in each round to ensure different bits are used.
- **PC-2:** Compressing 56 bits into a 48-bit subkey.

6 Java Concept: Bitwise XOR

In Java, the XOR operation is performed using the ^ operator.

```java
public class DESXorTest {
    public static void main(String[] args) {
        long plaintext = 0x12345678L;
        long key = 0xABCDEF01L;

        // XORing data with the key
        long ciphertext = plaintext ^ key;

        // Reversing the operation (XOR is self-reversible)
        long original = ciphertext ^ key;

        System.out.println("Original: " + Long.toHexString(
           plaintext));
```

```
        System.out.println("Result: " + Long.toHexString(original
            ));
    }
}
```

7 Complexity Analysis

The 56-bit key length is the primary weakness of DES:

- **Key space:** $2^{56} \approx 7.2 \times 10^{16}$ combinations.
- **Status:** Broken. Modern hardware can brute-force a DES key in a few hours.
- **Successor:** AES (Advanced Encryption Standard) with keys up to 256 bits.

8 Exercises

1. Why is the S-box step considered the most important part of DES security?
2. Perform a bitwise XOR on `101010` and `011011`.
3. Explain why DES uses 16 rounds instead of just one or two.
4. (Coding) Implement a simple function in Java that performs the Expansion (E) step for a 4-bit input expanding to 6 bits.